

Math Model Reverse Logistics

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Sets and indices		
C	Set of Collection Centres	$c \in N$
D	Set of Dismantling Centres	$d \in D$
R	Set of Recycling Centres	$r \in R$

Parameters		
$Trans_{cost}$	Cost of Transportation by vehicles	Euros
FC_d	Fixed Cost of opening a dismantling facility	Euros
FC_r	Fixed Cost of opening a recycling facility	Euros
$Cap_{d,max}$	Maximum capacity of dismantling centre d	Tons
$Cap_{r,max}$	Maximum capacity of recycling centre r	Tons
$dist_{cd}$	Distance of paths from collection centre c to dismantling centre d	km
$dist_{dr}$	Distance of paths from dismantling centre d to recycling centre r	km
α	Factor to convert the Motors to usable parts	
β	Factor to convert the parts to recycled magnets	
S_c	Amount of weight of motors collected at collection centre c	Tons
M_t	Amount of magnets required per time period t	Tons
E_m	Emission Factor	
$TotalCap_{cost}$	Total Capacity cost per ton	Euros
γ	Required percentage of demand that needs to be satisfied by recycled magnets	

The mathematical formulation is:

$$\begin{aligned}
 \min Z = & Trans_{cost} \cdot \left[\sum_{c \in N} \sum_{d \in D} dist_{cd} \cdot F_{cd} \right. \\
 & + \sum_{d \in D} \sum_{r \in R} dist_{dr} \cdot F_{dr} \left. \right] \\
 & + \sum_{d \in D} FC_d \cdot Y_d + \sum_{r \in R} FC_r \cdot Y_r \\
 & + TotalCap_{cost} \cdot \left[\sum_{d \in D} Cap_d + \sum_{r \in R} Cap_r \right] \\
 & + E_m \cdot \left[\sum_{c \in C} \sum_{d \in D} F_{cd} \cdot dist_{cd} + \sum_{d \in D} \sum_{r \in R} F_{dr} \cdot dist_{dr} \right]
 \end{aligned} \tag{1}$$

Decision Variables		
Y_c	Binary variable if collection centre c is established	
Y_d	Binary variable if dismantling centre d is established	
Y_r	Binary variable if recycling centre r is established	
F_{cd}	Transported weight or units of material from collection centre c to dismantling centre d	Ton
F_{dr}	Transported weight of material from dismantling centre d to recycling centre r	Ton
Cap_d	Capacity of the Dismantling centre c	
Cap_r	Capacity of the Recycling centre c	
z_d	Linearized capacity of collection centre d	
z_r	Linearized capacity of recycling centre r	

Table 1: Decision variables

0.1 Constraints

0.1.1 1. Facility Capacity Constraints

Motor Supply from Countries

$$S_c \geq \sum_{d \in D} F_{cd} \quad \forall c \in C \quad (2)$$

Dismantling Centres

$$\sum_{c \in C} F_{cd} \leq z_d \quad \forall d \in D \quad (3)$$

$$z_d \leq Cap_{d,max} \cdot Y_d \quad \forall d \in D \quad (4)$$

$$z_d \leq Cap_d \quad \forall d \in D \quad (5)$$

$$z_d \geq Cap_d - Cap_{d,max} \cdot (1 - Y_d) \quad \forall d \in D \quad (6)$$

$$\sum_{r \in R} F_{dr} \leq z_d \quad \forall d \in D \quad (7)$$

Recycling Centres

$$z_r \leq Cap_{r,max} \cdot Y_r \quad \forall r \in R \quad (8)$$

$$z_r \leq Cap_r \quad \forall r \in R \quad (9)$$

$$z_r \geq Cap_r - Cap_{r,max} \cdot (1 - Y_r) \quad \forall r \in R \quad (10)$$

$$\sum_{d \in D} F_{dr} \leq z_r \quad \forall r \in R \quad (11)$$

$$\sum_{r \in R} F_{dr} = \alpha \cdot \sum_{c \in C} F_{cd} \quad \forall d \in D \quad (12)$$

$$\gamma \cdot M_t \leq \beta \cdot \sum_{d \in D} \sum_{r \in R} F_{dr} \quad (13)$$

0.2 Facility Activation Constraints

$$\sum_{d \in D} Y_d \geq 1 \quad (14)$$

$$\sum_{r \in R} Y_r \geq 1 \quad (15)$$

$$Cap_d \leq Cap_{d,max} \quad \forall d \in D \quad (16)$$

$$Cap_r \leq Cap_{r,max} \quad \forall r \in R \quad (17)$$

0.2.1 7. Binary and Non-Negativity Constraints

$$Y_d, Y_r \in \{0, 1\} \quad (18)$$

$$F_{cd}, F_{dr} \geq 0 \quad (19)$$

$$Cap_d, Cap_r \geq 0 \quad (20)$$

$$z_d, z_r \geq 0 \quad (21)$$